

Sheet 6 — Eigenvalues and eigenvectors

Numbering continues from Sheet 5.

99. Find the eigenvalues and eigenvectors, if any, of each of the following matrices and state the algebraic and geometric multiplicities of each eigenvalue found.

$$\text{a) } \begin{bmatrix} 1 & 3 \\ 2 & -4 \end{bmatrix} \quad \text{b) } \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad \text{c) } \begin{bmatrix} 3 & 4 \\ 5 & 2 \end{bmatrix} \quad \text{d) } \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{e) } \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\text{f) } \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix} \quad \text{g) } \begin{bmatrix} 5 & 6 & -3 \\ -1 & 0 & 1 \\ 1 & 2 & -1 \end{bmatrix} \quad \text{h) } \begin{bmatrix} 2 & -1 & 2 \\ 5 & -3 & 3 \\ -1 & 0 & -2 \end{bmatrix} \quad \text{i) } \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\text{j) } \begin{bmatrix} 0 & 2 & 1 \\ -2 & 0 & 3 \\ -1 & -3 & 0 \end{bmatrix} \quad \text{k) } \begin{bmatrix} 2 & 5 & -6 \\ 4 & 6 & -9 \\ 3 & 6 & -8 \end{bmatrix} \quad \text{l) } \begin{bmatrix} 2 & 3 & 3 \\ 1 & 4 & 3 \\ -1 & -5 & -4 \end{bmatrix} \quad \text{m) } \begin{bmatrix} 1 & -1 & 2 \\ 3 & -1 & 0 \\ 1 & 2 & -1 \end{bmatrix}$$

$$\text{n) } \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{o) } \begin{bmatrix} 3 & -1 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & 5 \end{bmatrix}$$

$$\text{p) } \begin{bmatrix} 0 & 2 & -1 & -4 \\ 0 & 2 & 0 & -4 \\ 1 & -1 & -2 & 0 \\ -3 & 3 & 3 & -2 \end{bmatrix}$$

Item g) is more laborious: it has irrational eigenvalues.

100. Find the eigenvalues and eigenvectors of the linear operator $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ represented in the standard basis by the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 2 & 3 \end{bmatrix}.$$

101. Find the spectrum and the eigenvectors of the linear operator $G : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $G(x, y, z) = (y, -x, z)$.

102. Check whether the following matrices are diagonalisable. For those that are, find matrices P and D such that $P^{-1}AP = D$.

$$\text{a) } \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix} \quad \text{b) } \begin{bmatrix} 2 & 2 & -2 \\ 0 & 5 & -3 \\ 0 & 1 & 1 \end{bmatrix} \quad \text{c) } \begin{bmatrix} 0 & -2 & -4 \\ 0 & 3 & 5 \\ -2 & -3 & -3 \end{bmatrix} \quad \text{d) } \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\text{e) } \begin{bmatrix} -3 & -4 & 4 \\ 0 & -1 & 0 \\ -2 & -4 & 3 \end{bmatrix} \quad \text{f) } \begin{bmatrix} -1 & 3 & -1 \\ -4 & 7 & -2 \\ -8 & 10 & -2 \end{bmatrix} \quad \text{g) } \begin{bmatrix} 3 & -2 & 1 \\ 2 & -1 & 1 \\ -2 & 2 & 0 \end{bmatrix}$$

$$\text{h) } \begin{bmatrix} 4 & 0 & -4 & -3 \\ -2 & 1 & 3 & 2 \\ 2 & 0 & -2 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$